COMP 3512 - Group Tutorial #5 – March 27, 2026

Due on March 29, 2026 at 11:59 pm

You will receive a full, half, or zero mark depending on how much work you complete.

Please **disable AI assistance** in VS Code or Codespaces. These exercises are intended as practice for your exam, which will be written on paper and will not allow the use of AI tools. To disable AI assistance, follow the steps below:

- Open **VS Code** or **Codespaces**.
- Go to **File > Preferences > Settings** (or press **Ctrl + ,**).
- Search for **"disable AI"**.
- Locate **Chat** and disable settings related to **GitHub Copilot Chat**.
- You may need to refresh your Codespaces page or restart VS Code for the changes to take effect.

Group Work

In this tutorial, you will practice working with databases and superglobals in groups of two students. One of the main purposes of group tutorials is to help you learn how to collaborate on a project, resolve potential conflicts, and learn from and teach each other.

You may choose one of the following approaches:

- **Work on the entire assignment together**, which is recommended if you have questions or uncertainties about how to proceed. This approach allows you to discuss ideas, clarify concepts, and learn from one another.

- **Divide the tasks** (for example, implementing different functions) between group members and then combine your work. This approach is effective when both members clearly understand the assignment requirements and can work independently on smaller components.

You may also use a **hybrid approach**: begin by working together to understand the requirements, and once both group members feel confident about the implementation, divide the tasks. Choose the approach that best supports effective collaboration and learning for both group members.

This lab is expected to take approximately **40–50 minutes** to complete in a group setting.

Objective

In this assignment, you will build a simple **Single Page Application (SPA)** that displays and manages a list of books. You will start with a single page and gradually extend it into a two-view SPA using JSON data.

Submission

Share your work with me via GitHub or submit it through D2L **by Sunday March 29, 11:59pm**. Please include the names of both group members as a comment at the top of your main php files.

Instructions: Two-View SPA Using JSON

Tools: HTML and JavaScript only (no frameworks, no external libraries)

Part 1: Single View (Book List)

Create a webpage that displays a list of books. The starter code has been provided.

Requirements

1. Store given book data in **JSON format** (as a string).
2. Use **JSON.parse()** to convert the JSON data into a JavaScript object.
3. Display the list of books on the page using DOM manipulation.
4. Each book should display Title and Author

Expected Output:

Favorite Books

- 1984 by George Orwell
- To Kill a Mockingbird by Harper Lee
- The Great Gatsby by F. Scott Fitzgerald
- Pride and Prejudice by Jane Austen
- The Hobbit by J.R.R. Tolkien

Part 2: Two-View SPA (Add Book Feature)

Extend your application into a **two-view SPA**.

View 1: Book List

- Display all books.
- This view should update automatically when a new book is added.

View 2: Add Book

- Create a form to add a new book:
  - Title (input)
  - Author (input)
- When the form is submitted:
  - Add the new book to the existing data
  - Switch back to the list view
  - Update the displayed list

General Requirements

- Do **not** use inline event handlers (e.g., no onclick in HTML).
- Use addEventListener in JavaScript.
- All data should be handled on the **client side**.

Navigation

- Provide buttons to switch between the two views.
- Do **not reload the page** when switching views.

Optional

- Add a **Delete** button for each book.
- Use JSON.stringify() to convert your data back into JSON format after updates.

Expected Output

Favorite Books

View Books Add Book

Book List

- 1984 by George Orwell
- To Kill a Mockingbird by Harper Lee
- The Great Gatsby by F. Scott Fitzgerald
- Pride and Prejudice by Jane Austen
- The Hobbit by J.R.R. Tolkien

View 1: Book List - initially

Favorite Books

View Books Add Book

Add a New Book

Title: The Book of Healing
Author: Avicenna
Add Book

View 2: Add Book

Favorite Books

View Books Add Book

Book List

- 1984 by George Orwell
- To Kill a Mockingbird by Harper Lee
- The Great Gatsby by F. Scott Fitzgerald
- Pride and Prejudice by Jane Austen
- The Hobbit by J.R.R. Tolkien
- The Book of Healing by Avicenna

View1: Book List After Adding